

- (b) A diffraction grating is used with a laser as shown, and a line of bright spots is seen on the screen as shown.

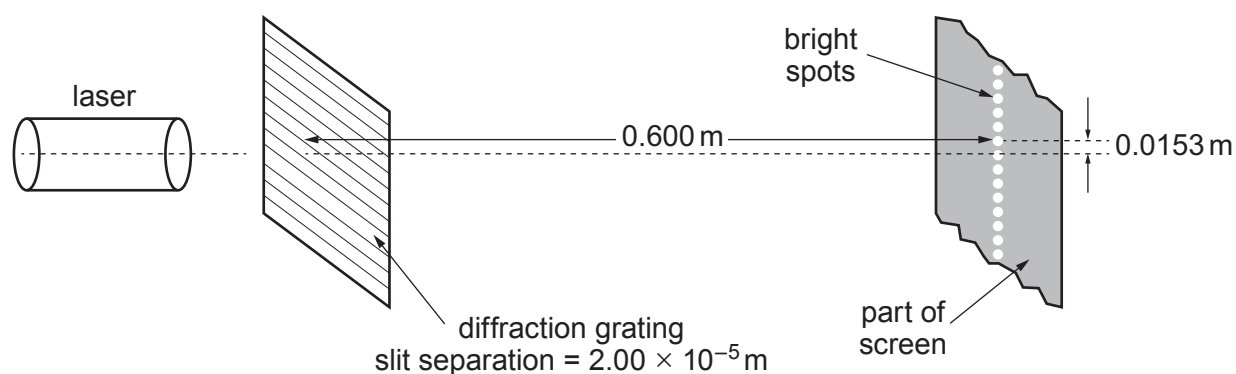


DIAGRAM NOT TO SCALE

- (i) Using a labelled sketch of a relevant triangle (in the space below), explain why, to a very good approximation:

$$\sin \theta_1 = \frac{0.0153}{0.600}$$

in which θ_1 is the angle between the zeroth order and first order beams.

[2]

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- (ii) Determine the wavelength of light from the laser.

[2]

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- (iii) Calculate the highest **order** produced by this grating for this wavelength, and suggest why so many orders are produced by this grating. [2]

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Examiner
only

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